

Data and Moments

Companion Specification for the Segmented Search Model

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1 Purpose of this document

This document specifies the data inputs, the targeted moments, and the moment-to-parameter mapping used to estimate the segmented search model. It is a companion to the model notes: the model notes define the equilibrium objects and the parameters; this document defines what is measured in the data, how each empirical moment is constructed, what it identifies, and how the empirical moments are combined into the SMM objective. It follows the structure: first the data sources and what we extract from each; then the moments and their identification role; then the estimation windows; then construction recipes; then the weight matrix, influence functions, and file layout; finally, the externally set parameters that are not part of the SMM loop. **Convention.** *Unskilled*: no completed four-year degree. *Skilled*: bachelor's

degree or higher. Formally, $\text{skilled}_i = \mathbf{1}\{\text{EDUC}_i \geq 111\}$ (IPUMS code 111 = Bachelor's degree). **Notation.** w_i denotes the survey weight (WTFINL_i for CPS Basic, ASECWT_i for ASEC).

$\text{LF}_i = \mathbf{1}\{\text{LABFORCE}_i = 2\}$; $\text{NILF}_i = 1 - \text{LF}_i$. $\text{train}_i = \mathbf{1}\{\text{SCHLCOLL}_i \in \{3, 4\} \text{ and } \text{EDUC}_i < 111\}$ (enrolled in college, no BA). $S_i = \mathbf{1}\{\text{EDUC}_i \geq 111\}$. Model objects: agg_uU , agg_eU are the unskilled unemployed/employed masses; agg_uS , agg_eS the skilled equivalents; agg_t the training mass; $\text{agg_mS} = \text{agg_uS} + \text{agg_eS}$; $\text{model_lf} = \text{agg_uU} + \text{agg_eU} + \text{agg_mS}$ (excludes the training mass); $\text{total_pop} = \text{model_lf} + \text{agg_t} = 1$. **Model variant covered.** The esti-

mation uses the variant with an exogenous skilled separation hazard $\xi_S \geq 0$ (skilled separations are $\xi_S + \lambda_S \Gamma(p_S^*)$), a discrete cross-market search policy $d(x) \in \{0, 1\}$ for skilled-unemployed workers, an aggregate production scale A entering both sectors, and log-normal wage measurement error $\sigma_{w,j}$ on the observed wage. In stationary equilibrium $d(x) \equiv 0$; the cross-market channel activates only along the simulated transition between equilibria. The SMM objective and the solver inside it work in stationary equilibrium, so the data-side construction proceeds under the convention $\text{EDUC} \geq 111 \Leftrightarrow \text{trained}$.

Part I Data Sources

2 Current Population Survey — Basic Monthly

What it gives us. Unemployment rates, the skilled share, the training share, the skilled long-term-unemployment share, all UE / EU transition rates, and the industry composition needed for the JOLTS vacancy allocation. The AGE variable also supports the life-table pre-estimation of ν , and the DURUNEMP variable supports the skilled long-term-unemployment share ltu_share_S .

Source. IPUMS-CPS Basic Monthly extracts (<https://cps.ipums.org>).

Extract window. January 2000 – December 2022 (with a one-month buffer for panel matching). This covers all four estimation windows defined in Part III.

Required variables.

Panel matching for transitions. The CPS uses a 4-8-4 rotation. Match on CPSIDP across consecutive months within a rotation spell. Valid month-pairs: $\text{MISH} \in \{1, 2, 3, 5, 6, 7\}$ matched to the following month. The same matched panel is used for the four CPS transition moments (jfr_j , sep_rate_j).

Table 1: CPS Basic Monthly — required variables.

IPUMS name	Type	Use
YEAR	id	Calendar year
MONTH	id	Calendar month
CPSID	id	Household linking key
CPSIDP	id	Person linking key (cross-month matching)
MISH	id	Month-in-sample (rotation group 1–8)
WTFINL	wt	Basic Monthly final weight
EMPSTAT	stat	Employment status (employed / unemployed / NILF)
LABFORCE	stat	In labour force (yes/no)
WHYABSNT	stat	Reason for absence (COVID misclassification correction)
DURUNEMP	stat	Weeks unemployed, current spell (continuous; NIU outside the unemployed universe). Long-term-unemployment flag for <code>ltu_share_S</code> : LTU iff $DURUNEMP \geq 27$.
EDUC	demog	Educational attainment (detailed). Skill indicator: $S = 1$ iff $EDUC \geq 111$.
SCHLCOLL	demog	Currently enrolled in school/college
AGE	demog	Age (restrict to 16–64; used in life-table ν)
SEX	demog	Sex
IND	occ	Industry (Census codes $\rightarrow$ 2-digit NAICS). Used for JOLTS vacancy allocation.
CLASSWKR	stat	Class of worker (wage/salary vs. self-employed)

Industry-to-supersector mapping. For the JOLTS vacancy allocation we map CPS IND (Census industry codes) to the 17 JOLTS supersectors used as the allocation basis (Section 4). The mapping is verified against the 2024 ACS code list and constructed so that each Census code routes to exactly one allocation supersector or is excluded (agriculture, armed forces, private households).

COVID-era correction. During March–June 2020, the CPS suffered response-rate drops and a well-documented misclassification of temporarily laid-off workers as “employed, absent from work.” We apply the BLS correction: for the March–June 2020 months, persons coded as employed with $WHYABSNT = 15$ (“other reason for absence”) are reclassified to $EMPSTAT = 21$ (unemployed on temporary layoff). This correction is applied before any moments are computed.

Codes used in the pipeline.

- *Employed*: $EMPSTAT \in \{10, 12\}$.
- *Unemployed*: $EMPSTAT \in \{20, 21, 22\}$.
- *NILF*: $EMPSTAT \in \{30, 31, 32, 33, 34, 36\}$.
- *Civilian labour force*: $LABFORCE = 2$.
- *Wage/salary worker*: $CLASSWKR \in \{22, 25, 27, 28\}$.

- *Enrolled in college without BA (the train flag):* SCHLCOLL $\in \{3, 4\}$ (full-time or part-time college) and EDUC < 111 .
- *Long-term unemployed (the LTU flag for `ltu_share_S`):* DURUNEMP ≥ 27 weeks, defined among the unemployed only (DURUNEMP is NIU for the employed and NILF, so the flag is evaluated on the skilled-unemployed stock).

3 CPS Annual Social and Economic Supplement (ASEC)

What it gives us. Wage moments: mean and median full-time-equivalent weekly earnings and the 25th percentile by skill group; wage dispersion (variance) and skewness (third central moment) by skill group; the log skill premium; and the cross-market wage overlap pair (`overlap_UgtS`, `overlap_SltU`). All wage moments are computed in *logs* on *real weekly-earnings levels* (no within-window normalisation; see Section 12).

Source. IPUMS-CPS ASEC samples.

Extract window. ASEC 2004–2023 (covering income earned in 2003–2022). This range covers all four estimation windows.

Table 2: CPS ASEC — required variables.

IPUMS name	Type	Use
YEAR	id	Survey year
CPSID	id	Household linking
CPSIDP	id	Person linking
ASECWT	wt	ASEC supplement weight
AGE	demog	Age (restrict 16–64)
SEX	demog	Sex
EDUC	demog	Educational attainment
EMPSTAT	stat	Employment status
INCWAGE	income	Annual wage/salary income
WKSWORK1	income	Weeks worked last year (continuous)
UHRSWORKLY	income	Usual hours per week last year
CLASSWKR	stat	Class of worker
CPI99	price	CPI-U deflator (or deflate with BLS CPI-U-RS)

Required variables.

Full-time-equivalent weekly-earnings construction.

$$w_{it} = \frac{\text{INCWAGE}_{it}}{\text{WKSWORK1}_{it} \times \text{UHRSWORKLY}_{it}} \times \bar{h} \times \frac{\text{CPI-U-RS}_{2012}}{\text{CPI-U-RS}_t}, \quad (1)$$

the real hourly wage scaled to a standard full-time week $\bar{h} = 40$ hours. Pricing every worker at a common full-time week removes part-time hours variation, so the wage distribution reflects pay rates rather than hours and matches the model’s hours-free wage. Trim at the 1st and 99th percentiles of the pooled within-window distribution. No median normalisation is applied: wage levels are matched in full-time-equivalent weekly units, with the model’s aggregate production scale A absorbing the dollar level (Section 12).

Sample restrictions (ASEC).

1. Age 16–64 at survey date.
2. Employed wage/salary workers (CLASSWKR).
3. Positive wage income, weeks, and hours.
4. Exclude self-employed and unpaid family workers.

4 JOLTS

What it gives us. Vacancy counts by industry, used together with CPS-based unemployment by skill to construct skilled and unskilled market tightness.

Source. BLS JOLTS flat files (job openings level, monthly, seasonally adjusted).

Extract window. December 2000 – December 2022. Covers the longest estimation window (crisis_covid) through 2022-12.

Required series. All series are seasonally adjusted levels (SA, level). The 17 industry series below are mutually exclusive and collectively exhaustive over nonfarm employment; they are used for the skill-share allocation. Agriculture (NAICS 11) is excluded because JOLTS covers nonfarm payrolls only. The six aggregate series are downloaded for cross-checking only and are *never* included in the allocation sum.

Series ID	Supersector	NAICS	Role
<i>Allocation series (sum these)</i>			
JTS1000000000000000JOL	Mining and logging	21	allocation
JTS2300000000000000JOL	Construction	23	allocation
JTS3200000000000000JOL	Durable goods manufacturing	31–33 ^a	allocation
JTS3400000000000000JOL	Nondurable goods manufacturing	31–33 ^a	allocation
JTS4200000000000000JOL	Wholesale trade	42	allocation
JTS4400000000000000JOL	Retail trade	44–45	allocation
JTS4800990000000000JOL	Transportation, warehousing & utilities	22, 48–49	allocation
JTS5100000000000000JOL	Information	51	allocation
JTS5100990000000000JOL	Financial activities	52–53	allocation
JTS5400990000000000JOL	Professional and business services	54–56	allocation
JTS6100000000000000JOL	Educational services	61	allocation
JTS6200000000000000JOL	Health care and social assistance	62	allocation
JTS7100000000000000JOL	Arts, entertainment, and recreation	71	allocation
JTS7200000000000000JOL	Accommodation and food services	72	allocation
JTS8100000000000000JOL	Other services	81	allocation
JTS9100000000000000JOL	Federal government	92	allocation
JTS9200000000000000JOL	State and local government	92	allocation
<i>Aggregate cross-checks only (do not sum)</i>			
JTS0000000000000000JOL	Total nonfarm	—	cross-check
JTS3000000000000000JOL	Manufacturing	31–33	cross-check
JTS4000000000000000JOL	Trade, transport & util.	22, 42, 44–45, 48–49	cross-check
JTS6000000000000000JOL	Education & health	61–62	cross-check
JTS7000000000000000JOL	Leisure & hospitality	71–72	cross-check
JTS9000000000000000JOL	Government	92	cross-check

^a Durable and nondurable goods manufacturing together constitute NAICS 31–33; the two series are mutually exclusive and sum to JTS3000000000000000JOL.

5 Census Job-to-Job Flows (J2J)

What it gives us. The skilled employer-to-employer rate `ee_rate_S`. J2J is preferred to the CPS employer-change question because (i) it has full coverage from 2000 onward, (ii) the employer change is identified from administrative LEHD records rather than a self-reported survey item, and (iii) it provides counts by education group (E0–E4) at quarterly frequency.

Source. Census Bureau J2J Origin-Destination flows, national-level public-use release (https://lehd.ces.census.gov/data/j2j/latest_release/us/).

Extract window. Go for latest release.

Series selection. Filter the national release with the following exact conditions, applied identically across all four windows:

- Seasonally adjusted flag `seasonadj` = *S*;
- Geography `geo_level` = *N* (national);
- Industry `industry` = 00 (all);
- All demographic margins set to “all”: `sex` = 0, `agegrp` = *A00*, `race` = *A0*, `ethnicity` = *A0*, `firmage` = 0, `firmsize` = 0;
- Education `education` = *E4* (Bachelor’s degree or higher) at the origin firm.

The relevant magnitudes are:

- `EEHire`: count of EE hires from origin firm *A* to destination firm *B* in quarter *t* (the J2J flow).
- `MainB`: stock of main-job-holding workers at firm *B*.

Quarterly EE rate and monthly conversion. For each retained row in window *w* the quarterly hazard is

$$ee_t^Q = \frac{EEHire_t}{MainB_t},$$

and the implied constant-hazard monthly rate is

$$ee_t^M = 1 - (1 - ee_t^Q)^{1/3}.$$

`ee_rate_S` is the unweighted mean of ee_t^M over the quarters falling within window *w*. If the E4 cell is missing in any quarter, that quarter is dropped and the omission is recorded in the diagnostics. There is no fallback to all-workers cells.

6 National Student Clearinghouse (NSC) + IPEDS

What it gives us. Two products from the same NSC file: (i) the ingredients for the training completion rate ϕ — average enrollment in 4-year and 2-year institutions (E_{4yr}, E_{2yr}), pooled across all available Fall semesters, plus the corresponding median time-to-degree (d_{4yr}, d_{2yr}); and (ii) the per-window NSC IPEDS-Universe total enrolment count that anchors the `training_share` level adjustment κ_w (Sections 10 and 15). NSC is used only as a deterministic level reference; it is not introduced as its own moment in the SMM vector (NSC has 3–5 Fall observations per window, which is too few to support an influence-function variance estimate).

Source. NSC Enrollment Coverage Report (annual, Excel), which provides IPEDS Universe enrollment counts for 4-year and 2-year institutions at the US level for each Fall semester. The file covers Fall 2003 – Fall 2024.

Sheet layout. The relevant sheet has rows indexed by (*State or Region, Institution Sector*) and one column per Fall semester. Column headers vary across years (e.g. `Fall_2003` in the earlier years, `Fall_2017` in later years, and `Fall_2024*` for preliminary estimates); the parser accepts all three formats. The US-level rows for “All 4-year Institutions” and “All 2-year Institutions” supply $E_{4\text{yr}}$ and $E_{2\text{yr}}$. The US-level row for “Overall” supplies the NSC IPEDS-Universe total count entering κ_w .

Time-to-degree values. NCES median time-to-degree

$$d_{4\text{yr}} = 49 \text{ months}, \quad d_{2\text{yr}} = 37 \text{ months}.$$

The 49-month figure for bachelor’s programmes captures the typical 4-year programme plus a small completion-lag premium; the 37-month figure for associate’s programmes likewise exceeds the nominal two-year length.

Why two programme types. Both 4-year (bachelor’s) and 2-year (associate’s) programmes constitute “training” in the model, since ϕ governs aggregate transition from training to skilled employment. Treating ϕ as the enrollment-weighted average across the two completion technologies (rather than picking one) avoids under-counting completion among workers in shorter programmes.

Part II Moments and Identification

7 Moment catalogue

Table 3 lists all 26 targeted moments. Each is computed separately for every estimation window. The first 23 are the original block; the final three — the skilled long-term-unemployment share `ltu_share_S` and the cross-market wage-overlap pair (`overlap_UgtS`, `overlap_SltU`) — are appended at the end so the legacy moment indices and the data-side $\hat{\Sigma}$ columns are unchanged.

Table 3: Targeted moments.

Code name	Block	Definition
<code>ur_total</code>	Stocks	Overall unemployment rate, on the training-excluded LF denominator: $(U_U + U_S)/(N_U + N_S)$ with $\text{train}_i = 0$ enforced on the denominator.
<code>ur_U</code>	Stocks	Unskilled unemployment rate: U_U/N_U , denominator restricted to $\text{LF} \cap \neg\text{train}$.
<code>ur_S</code>	Stocks	Skilled unemployment rate: U_S/N_S .
<code>skilled_share</code>	Stocks	Skilled share of training-excluded LF: $N_S/(N_U + N_S)$ with $\text{train}_i = 0$ enforced on the denominator.
<code>training_share</code>	Stocks	Full-time students (enrolled in college without BA and NILF) as a fraction of working-age (16–64) population. Level is rescaled to the NSC IPEDS-Universe anchor via κ_w (Section 10).

Code name	Block	Definition
emp_var_U	Wage shape	Variance of log real weekly earnings, employed unskilled. Includes the measurement-error component $\sigma_{w,U}^2$.
emp_cm3_U	Wage shape	Third central moment (skewness) of log real weekly earnings, employed unskilled. Invariant to symmetric measurement error.
emp_var_S	Wage shape	Variance of log real weekly earnings, employed skilled. Includes $\sigma_{w,S}^2$.
emp_cm3_S	Wage shape	Third central moment (skewness) of log real weekly earnings, employed skilled.
jfr_U	Trans	Monthly unskilled job-finding rate: $\Pr(E_{t+1} U_t, U)$.
sep_rate_U	Trans	Monthly unskilled EU separation rate: $\Pr(U_{t+1} E_t, U)$.
jfr_S	Trans	Monthly skilled job-finding rate: $\Pr(E_{t+1} U_t, S)$.
sep_rate_S	Trans	Monthly skilled EU separation rate: $\Pr(U_{t+1} E_t, S)$.
ee_rate_S	Trans	Monthly skilled employer-to-employer rate (Census J2J, E4 only).
mean_wage_U	Wages	Mean log real weekly earnings, employed unskilled.
mean_wage_S	Wages	Mean log real weekly earnings, employed skilled.
p25_wage_U	Wages	25th-percentile log real weekly earnings, employed unskilled.
p25_wage_S	Wages	25th-percentile log real weekly earnings, employed skilled.
p50_wage_U	Wages	Median log real weekly earnings, employed unskilled.
p50_wage_S	Wages	Median log real weekly earnings, employed skilled.
wage_premium	Wages	Log skill premium: $E[\log w S] - E[\log w U]$.
theta_U	Tight	Unskilled V-U ratio: $\theta_U = V_U/U_U$.
theta_S	Tight	Skilled V-U ratio: $\theta_S = V_S/U_S$.
ltu_share_S	Duration	Skilled long-term-unemployment share: WTFINL-weighted share of the skilled unemployed whose current spell is long-term ($\text{DURUNEMP} \geq 27$ weeks), computed per month and averaged over the window. Model counterpart: the u_S -weighted survivor of the skilled unemployment spell past $a^* = 27 \cdot 12/52 \approx 6.23$ months under the exit hazard $\delta_S(x) = \kappa_S(1 - \Gamma(p_S^*(x))) + \nu$ (job-finding plus demographic exit).

Code name	Block	Definition
overlap_UgtS	Overlap	Cross-market wage overlap: ASECWT-weighted share of employed unskilled with log earnings above the skilled log-median, $\Pr(\log w_U > \text{med } \log w_S)$. Per ASEC year, then averaged.
overlap_SltU	Overlap	Cross-market wage overlap: ASECWT-weighted share of employed skilled with log earnings below the unskilled log-median, $\Pr(\log w_S < \text{med } \log w_U)$. Per ASEC year, then averaged.

8 Identification

This section explains *what each moment identifies* in the model: which parameters each moment primarily constrains, and why the set as a whole is sufficient.

8.1 Stock moments

- **ur_U, ur_S**: pin down the balance between job-finding and separation rates in each segment. Given tightness, they constrain matching efficiencies and shock hazards.
- **skilled_share**: constrains the training cutoff $\bar{x}$ at the long-run-stock margin and therefore the interaction between the type distribution (a_ℓ, b_ℓ) and the training cost coefficient c .
- **training_share**: identifies the mass in the training state, hence the cutoff $\bar{x}$ conditional on the (pre-estimated) training-completion rate ϕ .

8.2 Wage shape

- **emp_var_U**: variance of the unskilled log-wage distribution. Because the unskilled wage is linear in match quality p for fixed x (see model notes), the structural variance of wages maps into the shape of the match-quality distribution $G(p) = p^{\alpha_U}$ and into the regime-specific bargaining weight β_U , which scales the pass-through of match-quality heterogeneity into wages. The *observed* variance adds the measurement-error term $\sigma_{w,U}^2$, so **emp_var_U** jointly identifies the structural dispersion and $\sigma_{w,U}$: the structural shape is pinned by the transition and skewness moments, and $\sigma_{w,U}$ absorbs the residual cross-sectional log-wage variance the model cannot generate at plausible β_U . It is identified jointly with the reservation cutoff p^* (pinned independently by **sep_rate_U**).
- **emp_var_S**: analogous role for the skilled block. The structural part constrains the offer distribution $\Gamma = \text{Beta}(a_\Gamma, b_\Gamma)$, modulated by the skill-bias exponent γ_{P_S} via the convex transformation $x \mapsto x^{\gamma_{P_S}}$, and scaled by β_S ; the residual is absorbed by $\sigma_{w,S}$.
- **emp_cm3_U, emp_cm3_S**: the third central moment (skewness) of the employed log-wage distribution. Because the measurement error is symmetric (Gaussian in logs), the third central moment is *invariant* to $\sigma_{w,j}$ and is therefore a clean structural moment. The variance fixes the spread but not the asymmetry; the third moment identifies the *shape* of the match-quality / offer distribution beyond its scale — the damage-shape α_U (via $G(p) = p^{\alpha_U}$) in the unskilled block and the offer shape (a_Γ, b_Γ) in the skilled block — separately from the bargaining weight β_j , which rescales the whole distribution without changing its skew. Variance and skewness jointly therefore pin the shape parameters apart from the bargaining weights and the measurement error.

8.3 Transition rates

- **jfr_U**: in the model the unskilled job-finding rate equals the contact rate $f_U = \mu_U \theta_U^{1-\eta_U}$, because all new unskilled matches start at frontier efficiency $p_0 = 1 > p^*(x)$, so acceptance is automatic. Given θ_U from JOLTS, **jfr_U** identifies the regime-specific unskilled matching parameters (μ_U, η_U) .
- **jfr_S**: the skilled job-finding rate is the acceptance-weighted contact rate. Because a skilled offer is a draw $\tilde{p} \sim \Gamma$ and the worker rejects whenever $\tilde{p} < p_S^*(x)$, the model object is

$$\text{jfr_S}^{\text{model}} = \frac{\kappa_S \int (1 - \Gamma(p_S^*(x))) u_S(x) dx}{\int u_S(x) dx}, \quad \kappa_S = \mu_S \theta_S^{1-\eta_S}.$$

Given θ_S it identifies (μ_S, η_S) jointly with the lower tail of the offer distribution.

- **sep_rate_U**, **sep_rate_S**: the EU flow. In the unskilled block the model object is the employment-weighted endogenous destruction hazard

$$\text{sep_rate_U}^{\text{model}} = \frac{\lambda_U \int G(p^*(x)) e_U(x) dx}{\int e_U(x) dx}, \quad G(p) = p^{\alpha_U}.$$

In the skilled block, with the exogenous hazard ξ_S , the model object is the total destruction flow

$$\text{sep_rate_S}^{\text{model}} = \frac{\int (\xi_S + \lambda_S \Gamma(p_S^*(x))) e_S^{\text{tot}}(x) dx}{\int e_S^{\text{tot}}(x) dx}.$$

sep_rate_S therefore measures the *total* skilled separation hazard $\xi_S + \lambda_S \Gamma(p_S^*)$; the endogenous part is pinned by the wage moments through p_S^* , so ξ_S is identified as the residual separation flow that the endogenous channel cannot generate (it is the binding component once $p_S^*(x)$ falls toward zero). Demographic exits (rate ν) appear on neither side: in the data they map to E→NILF transitions, in the model they exit the population. Jointly with the wage-variance moments, these moments identify the shock arrival rates and the reservation cutoffs.

- **ee_rate_S**: the skilled job-to-job rate. In the model this is the OJS quit hazard

$$\text{ee_rate_S}^{\text{model}} = \frac{\int s^*(x, p) \kappa_S (1 - \Gamma(p)) e_S(x, p) dx dp}{\int e_S^{\text{tot}}(x) dx},$$

where $s^*(x, p) = \mathbf{1}\{p < p_S^{\text{oj}}(x)\}$ is the on-the-job search policy. This is the key moment separating the offer distribution Γ from the separation cutoff p_S^* , and also pins down the OJS cost σ_S (since workers search only if the ladder gain exceeds σ_S).

8.4 Wages

- **mean_wage_U**, **mean_wage_S**, **p50_wage_U**, **p50_wage_S**: mean and median log real weekly earnings by skill group. Wage *levels* identify the aggregate production scale A and the productivity shifters P_U and γ_{P_S} given the regime-specific bargaining weights β_j . Concretely, the overall level pins A , while the two sectoral levels pin the products $A P_U$ (unskilled) and $A \gamma_{P_S}$ (skilled); A is separated from the relative productivities because the skilled wage *curvature* in x (governed by the exponent γ_{P_S} , which is the same parameter as the skilled coefficient) is observed in the skilled wage shape, so the skilled coefficient cannot be rescaled freely to absorb A . The median is robust to the upper tail of the (trimmed) ASEC distribution; the mean adds the upper-tail information the median discards, so the two together discipline the level jointly with the dispersion moments. Both mean and median in logs are invariant to symmetric measurement error.
- **p25_wage_U**, **p25_wage_S**: the 25th percentile pins the *lower tail* of the wage distribution, which in the model is populated by the worst-quality matches that survive (those just

above the destruction cutoff p^*). Together with the variance it locates the reservation cutoff p_j^* and the lower tail of the match-quality / offer distribution. Under measurement error the percentile is the $\sigma_{w,j}$ -convolved quantile.

- **wage_premium**: the log skill premium is the primary moment identifying the *return to training*. It constrains γ_{P_S} relative to P_U (the aggregate scale A cancels in the difference) and the value of the training option. Invariant to the overall wage scale and to symmetric measurement error, so it is the cleanest of the wage moments at the model-data interface.

8.5 Tightness

- **theta_U, theta_S**: directly constrain regime-specific vacancy costs $\tilde{k}_U, \tilde{k}_S$ through free entry. Given matching parameters and the composition of unemployed workers, free entry pins down tightness uniquely.

8.6 Duration and cross-market overlap

- **ltu_share_S**: the skilled long-term-unemployment share. In the model the skilled-unemployed spell of a type- x worker is exponential with constant exit hazard $\delta_S(x) = \kappa_S(1 - \Gamma(p_S^*(x))) + \nu$ — job-finding plus the demographic exit ν , the model's only route out of the labour force. The cross-sectional share with elapsed duration past $a^* = 27 \cdot 12/52 \approx 6.23$ months is the u_S -weighted mixture survivor $\int u_S(x) e^{-\delta_S(x) a^*} dx / \int u_S(x) dx$. Because **jfr_S** already pins the *mean* skilled job-finding hazard, **ltu_share_S** adds the *dispersion* of $\delta_S(x)$ across types: a fatter long-duration tail requires more heterogeneity in skilled job-finding (length-biased sampling makes low-hazard types accumulate in the stock). Because the demographic exit ν is common to all types, it is the moment that disciplines the spread of the skilled offer-acceptance hazard around the mean.
- **overlap_UgtS, overlap_SltU**: the cross-market wage overlap, one moment per bargaining weight. **overlap_UgtS** is the mass of the unskilled log-wage distribution lying above the skilled median; **overlap_SltU** is the mass of the skilled distribution lying below the unskilled median. Together they locate the two log-wage distributions relative to each other *beyond* the gap in means (**wage_premium**), so they sharpen identification of the bargaining weights β_U, β_S — which govern how much match-quality heterogeneity passes through into each segment's wage spread, and hence how far each distribution reaches into the other's support — jointly with the variance and skewness moments.

8.7 Deep vs. regime-specific identification

Under the current parameter classification, the deep set is $\{a_\ell, b_\ell, \tilde{b}_U, \tilde{b}_T, \tilde{b}_S\}$ (the flow values entering relative to the aggregate scale, $b_j = \tilde{b}_j A$). These parameters are identified primarily from stock moments and the skill premium, which respond slowly across regimes:

- The type distribution (a_ℓ, b_ℓ) is identified primarily from **skilled_share** and **wage_premium**, which together pin down the location and dispersion of innate ability.
- Flow payoffs $\tilde{b}_U, \tilde{b}_T, \tilde{b}_S$ are identified from the level of unemployment values and training incentives, manifested in unemployment rates, the skilled share, and the position of wage distributions near the reservation cutoffs.

All other parameters — the training cost coefficient c , the aggregate production scale A , the matching parameters (η_j, μ_j) , the bargaining weights β_j , the OJS cost σ_S , the productivity shifters (P_U, γ_{P_S}) , the vacancy costs $(\tilde{k}_U, \tilde{k}_S)$, the shock hazards (λ_j) , the exogenous skilled separation ξ_S , the match-quality shape parameters $(\alpha_U, a_\Gamma, b_\Gamma)$, and the measurement-error scales $(\sigma_{w,U}, \sigma_{w,S})$ — are regime-specific and identified *within each window*:

- Training cost coefficient c : from **training_share** (the stock in the training state) in conjunction with **skilled_share**, since c governs the level of the training cutoff $\bar{x}$. Given

the pre-estimated completion rate ϕ , the steady-state training stock maps to the entry cutoff; regime variation in `training_share` and `skilled_share` between `base_covid` and `crisis_covid` is informative about whether a COVID-era training change reflects a shift in c (the decision rule) or a compositional change in the unskilled-unemployed pool.

- Aggregate scale A : from the overall level of `mean_wage_j` and `p50_wage_j`, jointly with the productivity shifters (which the premium separates from A). A is the demand-shock carrier: a common fall in wage levels across both sectors at unchanged premium and dispersion is read by the model as a fall in A .
- Matching parameters (η_j, μ_j) : from the joint level of `jfr_j` and `theta_j` within window.
- Bargaining weights β_j : from wage dispersion (`emp_var_j`, net of $\sigma_{w,j}^2$) and the cross-market overlap (`overlap_UgtS`, `overlap_SltU`) within window.
- Measurement error $\sigma_{w,j}$: from `emp_var_j` as the residual log-wage variance not accounted for by the structural dispersion (which the skewness and transition moments pin down).
- OJS cost σ_S : from `ee_rate_S` jointly with the wage moments, since the OJS cutoff p_S^{oj} responds to σ_S holding match-quality shape fixed.
- Productivity shifters and vacancy costs: from wage levels and tightness levels.
- Shock arrival rates, the exogenous skilled hazard ξ_S , and damage / offer shapes: from separation rates, wage-variance / skewness shifts, and the skilled long-term-unemployment share (`ltu_share_S`), which constrains the dispersion of the skilled exit hazard.

Testable implication: did matching technology shift? A pure aggregate-demand shock with no technology change predicts that the matching elasticities, efficiencies, bargaining weights, and OJS cost are *approximately equal* across the base→crisis pair within a single crisis episode, with the shock loading on A (and to a lesser extent the productivity shifters and vacancy costs). A supply / reallocation shock that re-curves the Beveridge curve permits these parameters to differ. The two crisis pairs in this paper deliver opposite verdicts on this test: the FC pair is consistent with stable matching and bargaining technology (a textbook ride along the Beveridge curve, loading on A), whereas the COVID pair shows a large shift in separation rates at nearly unchanged tightness, which can only be rationalised by a change in some subset of $(\mu_j, \eta_j, \beta_j, \sigma_S, \lambda_j, \xi_S, \alpha_U, \Gamma)$. The multi-regime estimation recovers the answer.

Part III Estimation Windows

9 The four windows

The multi-regime estimation requires moments computed over four distinct periods, each approximating a stationary regime: Each window produces 26 moment values plus a 26×26

Table 4: Estimation windows.

Label	Code	Period	Role
Pre-FC baseline	<code>base_fc</code>	Jan 2003 – Dec 2007	Stage 1: estimate all parameters (60 months).
Financial crisis	<code>crisis_fc</code>	Jan 2008 – Dec 2010	Stage 2a: regime-specific only (36 months).
Pre-COVID baseline	<code>base_covid</code>	Jan 2015 – Dec 2019	Structural stability check + 2nd baseline (60 months).
COVID crisis	<code>crisis_covid</code>	Jan 2020 – Dec 2022	Stage 2b: regime-specific only (36 months).

variance-covariance matrix from influence functions. The two baseline windows are 60 months and the two crisis windows are 36 months, by symmetric design.

9.1 Common sample restrictions

1. Age 16–64, civilian labour force.
2. Exclude armed forces.
3. Wage moments: wage/salary workers, positive income/weeks/hours, exclude self-employed.
4. Transition rates: matched month-pairs with valid CPSIDP, MISH $\in \{1, 2, 3, 5, 6, 7\}$.

9.2 Window-specific notes

base_fc (Jan 2003 – Dec 2007). Starts roughly a year after the 2002 trough of the post-dot-com recession. NBER expansion: Nov 2001 – Dec 2007.

crisis_fc (Jan 2008 – Dec 2010). Covers the NBER contraction (Dec 2007 – Jun 2009) and the slow post-trough recovery through 2010, when tightness had fallen sharply and had not yet returned to the pre-FC locus. The window ends before the 2011 fiscal cliff and the slow normalisation that followed.

base_covid (Jan 2015 – Dec 2019). Mature post-crisis expansion. Starting in 2015 avoids the early recovery (2010–2014). Tests structural stability of deep parameters across a decade.

crisis_covid (Jan 2020 – Dec 2022). NBER peak Feb 2020; the contraction was officially short, but the labour market did not return to the pre-COVID locus by end-2022. The window covers the pandemic shock, the rebound, and the sustained outward Beveridge-curve shift through 2022. BLS misclassification correction applied for Mar–Jun 2020.

9.3 Symmetric window design

The two baseline windows are 60 months each and the two crisis windows are 36 months each. This symmetric design lets the SMM treat the two crisis pairs on equal footing. The 36-month length is long enough to span the bulk of each crisis episode plus the immediate adjustment, and short enough to remain a distinguishable regime from the surrounding baseline.

FC: a ride along the Beveridge curve. Between **base_fc** and **crisis_fc** market tightness fell sharply (by roughly a third), unemployment rose, and separation rates rose modestly. The Beveridge curve traced out by the time series over 2008–2010 lies approximately on the same locus as the pre-FC data. This is the textbook signature of a pure aggregate-demand shock with unchanged matching, bargaining, and OJS technology.

COVID: an outward Beveridge-curve shift. Between **base_covid** and **crisis_covid** the Beveridge curve shifted outward: tightness fell only modestly, but separation rates roughly doubled for both skill groups. The pre-shock locus is not recovered by end-2022. This is consistent with a change in some subset of the matching / bargaining / OJS / shock parameters, which the model classifies as regime-specific precisely so that the data can answer the question.

9.4 ASEC year mapping

Since ASEC reports income for the *previous* calendar year, the ASEC surveys used for each window are:

- **base_fc:** ASEC 2004–2008 (income years 2003–2007).

- `crisis_fc`: ASEC 2009–2011 (income years 2008–2010).
- `base_covid`: ASEC 2016–2020 (income years 2015–2019).
- `crisis_covid`: ASEC 2021–2023 (income years 2020–2022).

9.5 Stationarity caveat

The model produces stationary-equilibrium moments. Short crisis windows are not true stationary states. We interpret crisis-window averages as the “quasi-stationary” centre of gravity. In stationary equilibrium the discrete cross-market policy $d(x)$ is identically zero; it activates only along the simulated transition between equilibria.

Part IV Moment Construction Recipes

10 Labour-market stocks

All recipes below are applied identically within each window w . Stock moments are computed by the *per-period-then-average* approach: compute a monthly value first, then average across all months in the window. All four stock denominators that touch the labour force enforce the working-student exclusion $LF_i \wedge \neg \text{train}_i$ so that the data denominator aligns with the model’s `model_lf`, which excludes the training mass.

ur_total. For each month t in window w :

$$\text{ur_total}_t = \frac{\sum_{i: \text{unemp}_t} w_i}{\sum_{i: LF_t \wedge \neg \text{train}_t} w_i}.$$

Average over all months in the window. The denominator excludes working students; the numerator is unaffected because employed absent-from-work students and unemployed students are exclusive categories under the IPUMS coding.

ur_U.

$$\text{ur_U}_t = \frac{\sum_{i: \text{unemp}_t, \neg S} w_i}{\sum_{i: LF_t \wedge \neg \text{train}_t, \neg S} w_i}.$$

Same working-student filter on the denominator as for **ur_total**, restricted to unskilled ($S = 0$).

ur_S.

$$\text{ur_S}_t = \frac{\sum_{i: \text{unemp}_t, S} w_i}{\sum_{i: LF_t, S} w_i}.$$

No working-student filter needed: by construction $\text{train} = 1 \Rightarrow \text{EDUC} < 111$, so the skilled denominator never contains a trainee.

skilled_share. Monthly fraction of the training-excluded LF that is skilled, averaged over the window:

$$\text{skilled_share}_w = \frac{1}{T_w} \sum_{t \in w} \frac{\sum_{i: LF_t, S} w_{i,t}}{\sum_{i: LF_t \wedge \neg \text{train}_t} w_{i,t}}.$$

The denominator carries the working-student filter; the numerator does not (working students are by construction not skilled).

training_share (raw). Strict variant: the numerator is restricted to NILF trainees (full-time students whose state coincides with the model’s training state, which sits outside the labour force), and the denominator is the working-age population (already restricted to 16–64 upstream):

$$\text{training_share}_w = \frac{1}{T_w} \sum_{t \in w} \frac{\sum_{i: \text{train}_t \wedge \text{NILF}_t} w_{i,t}}{\sum_{i: \text{pop}_t} w_{i,t}}.$$

This matches the model definition $\text{training_share} = \text{agg_t}/\text{total_pop}$ with $\text{total_pop} = 1$. The raw level produced by this expression is under-coverage-biased and not cross-window comparable; the next paragraphs introduce the NSC-anchored adjustment that the SMM actually targets.

ltu_share_S (skilled long-term-unemployment share). WTFINL-weighted share of the skilled unemployed whose current spell is long-term ($\text{DURUNEMP} \geq 27$ weeks), computed per month and averaged across the window:

$$\text{ltu_share_S}_w = \frac{1}{T_w} \sum_{t \in w} \frac{\sum_{i: \text{unemp}_t, S, \text{DURUNEMP}_{i,t} \geq 27} w_{i,t}}{\sum_{i: \text{unemp}_t, S} w_{i,t}}.$$

Cross-sectional — no panel matching: the denominator is the current skilled-unemployed stock and the numerator its long-term subset. Months with no skilled unemployed are skipped before averaging. The model counterpart is the u_S -weighted survivor past $a^* = 27 \cdot 12/52 \approx 6.23$ months (Section 8.6); the only competing risk that enters that survivor is the demographic exit ν , common to all types and pre-estimated from the life table, not from DURUNEMP .

Universe note for training_share. IPUMS SCHLCOLL is collected only for ages 16–24 in the pre-2013 era and is universe-expanded to ages 16–54 from Jan 2013 onward. Within `base_fc` and `crisis_fc` the universe is consistent (both restrict to 16–24 implicitly through the missing-data structure). Within `base_covid` and `crisis_covid` the universe is also consistent (both include 25+). Cross-FC-vs-COVID comparisons of the raw CPS moment, however, are confounded by the universe change. A direct diagnostic is to tabulate the CPS `in_training` share by age band $\times$ window: the bands 25+ are structurally zero in the FC windows and non-trivial in the COVID windows, precisely because of the SCHLCOLL universe limit.

NSC-based level adjustment (κ_w). To make `training_share` comparable across the FC and COVID windows and to correct its absolute level (the raw CPS construction understates the true postsecondary trainee mass even within each window), we apply a per-window deterministic scale factor calibrated to the NSC IPEDS-Universe Fall headcount. Let $\widehat{\text{enr}}_w^{\text{NSC}}$ be the NSC US-“Overall” IPEDS Universe enrolment, averaged across the Fall semesters falling within window w , and let $\widehat{\text{enr}}_w^{\text{CPS}} = \frac{1}{T_w} \sum_{t \in w} \sum_{i: \text{train}_t \wedge \text{NILF}_t} w_{i,t}$ be the CPS-implied enrolment count (per-month then averaged, identical numerator weighting to the raw moment). Define

$$\kappa_w = \frac{\widehat{\text{enr}}_w^{\text{NSC}}}{\widehat{\text{enr}}_w^{\text{CPS}}}.$$

The adjusted data target consumed by the SMM is

$$\text{training_share}_w^{\text{adj}} = \kappa_w \cdot \text{training_share}_w \equiv \frac{\widehat{\text{enr}}_w^{\text{NSC}}}{\widehat{\text{Pop}}_w^{\text{CPS}}}.$$

The model counterpart $\text{agg_t}/\text{total_pop}$ is age-uncapped by construction and is directly comparable to the adjusted target. This adjustment is applied *in the data pipeline* (Stage 7 writes the κ_w -scaled moment to `moments_{window}.csv`), so the SMM loaders read the pre-adjusted target; see Sections 14 and 15.

Identifying assumptions for κ_w . The rescaling is exact under three assumptions. Posit a multiplicative coverage model

$$\begin{aligned}\hat{m}_w^{\text{CPS}} &= \alpha_w \tilde{m}_w (1 + \varepsilon_w^{\text{CPS}}), \\ \hat{m}_w^{\text{NSC}} &= \tilde{m}_w (1 + \varepsilon_w^{\text{NSC}}),\end{aligned}$$

where $\tilde{m}_w$ is the true age-uncapped postsecondary training share, $\alpha_w < 1$ captures the CPS under-coverage, and the ε 's are mean-zero noise.

- **A1 (NSC $\approx$ census).** $\text{Var}(\hat{m}_w^{\text{NSC}}) \approx 0$. NSC is an institutional reporting system, not a survey; the IPEDS Universe coverage rate is published in the Coverage sheet of the same NSC file and runs $\approx 86\%$ in 2003 climbing to $\approx 98\%$ from 2015 onward. Residual non-coverage is institutional (small or specialty schools that don't report), not random sampling, so it does not contribute a variance term.
- **A2 (α_w is structural within window).** α_w is a fixed property of the CPS measurement instrument in window w , not a behavioural object. Three components support this: (i) the SCHLCOLL age universe is a hard IPUMS feature and changes only in Jan 2013, between the FC and COVID pairs; (ii) the EDUC < 111 filter is definitional; (iii) the NILF intersection is part of the model-data mapping (the model's training state is defined as NILF), not a measurement choice that worker behaviour can invalidate.
- **A3 (Plug-in for $\hat{\kappa}_w$).** Treat the realised $\hat{\kappa}_w$ as constant. This is the standard two-step estimator move. Here it biases the variance *upward*: the product $\hat{\kappa}_w \cdot \hat{m}_w^{\text{CPS}}$ collapses algebraically to $\widehat{\text{enr}}_w^{\text{NSC}} / \widehat{\text{Pop}}_w^{\text{CPS}}$, and the CPS population total has CV well under 1%, so the true $\text{Var}(\hat{m}_w^{\text{adj}})$ is substantially smaller than $\kappa_w^2 \text{Var}(\hat{m}_w^{\text{CPS}})$. The SMM therefore down-weights `training_share` *more* than the optimal-weight matrix would, which is a loss of efficiency, not a bias.

Under A1–A3, bilinearity gives exactly

$$\text{Var}(\hat{m}_w^{\text{adj}}) = \kappa_w^2 \text{Var}(\hat{m}_w^{\text{CPS}}), \quad \text{Cov}(\hat{m}_w^{\text{adj}}, m_k) = \kappa_w \text{Cov}(\hat{m}_w^{\text{CPS}}, m_k) \quad \forall k.$$

Off-diagonals are *preserved up to κ_w* and are not zeroed, so the full CPS covariance structure of the other 25 moments is unchanged.

Grad-student note. NSC IPEDS-Universe counts all Title-IV postsecondary enrolment, including graduate students, while CPS `in_training` excludes them via EDUC ≥ 111 . κ_w therefore absorbs the grad-student share ($\approx 13\%$ of total US postsecondary enrolment) on top of the SCHLCOLL universe correction. The model's training state, interpreted as “investment in becoming skilled,” covers grad school in the current variant, so this interpretation is consistent with the SMM target.

emp_var_U, emp_var_S, emp_cm3_U, emp_cm3_S. ASECWT-weighted variance and third central moment of the log real weekly earnings $\log w$ (Section 12) by skill group, computed within each ASEC year and averaged across years in the window. The variance carries the measurement-error component $\sigma_{w,j}^2$; the third central moment is invariant to symmetric measurement error.

11 Transition rates

jfr_U, jfr_S. From matched CPS month-pairs $(t, t+1)$ in window w :

$$\text{jfr}_{j,t} = \frac{\sum_{i: U_t \rightarrow E_{t+1}, j} w_{i,t}}{\sum_{i: U_t, j} w_{i,t}}, \quad j \in \{U, S\}.$$

Averaged over all valid month-pairs in the window. Weight: base-month WTFINL.

sep_rate_U, sep_rate_S.

$$\text{sep}_{j,t} = \frac{\sum_{i: E_t \rightarrow U_{t+1}, j} w_{i,t}}{\sum_{i: E_t, j} w_{i,t}}.$$

This is the $E \rightarrow U$ flow only. Job-to-job moves are not separations. $E \rightarrow \text{NILF}$ transitions are also excluded by construction (they map to demographic exit, not unemployment). On the skilled side this empirical flow corresponds to the total model hazard $\xi_S + \lambda_S \Gamma(p_S^*)$.

Frequency consistency. The CPS transition moments above are monthly transition *probabilities*. The model reports rates as continuous-time Poisson *hazards* ($f = \mu\theta^{1-\eta}$, $\text{sep} = \lambda G(p^*)$, $\text{ee} = \kappa(1 - \Gamma)$), so for a monthly probability p the targeted moment is the implied hazard $h = -\log(1 - p)$. This is applied to **jfr_j**, **sep_rate_j** and **ee_rate_S**; stocks, ratios, wage moments, the overlaps, and **ltu_share_S** are not rates and are left as-is.

ee_rate_S (from J2J, E4 only). Apply the J2J filter from Section 5 restricting to **education** = E4 at origin. For each retained quarter t , the quarterly hazard is $\text{ee}_t^Q = \text{EEHire}_t / \text{MainB}_t$ and the implied monthly hazard is $\text{ee}_t^M = 1 - (1 - \text{ee}_t^Q)^{1/3}$. **ee_rate_S** is the unweighted mean of ee_t^M over the quarters falling within window w .

12 Wages

All from the ASEC. Real weekly earnings from (1), trimmed at 1st/99th percentiles within the window, then logged.

No normalisation: levels in real weekly earnings. Wages enter the moments as $\log w_{it}$ on real (CPI-deflated) weekly-earnings *levels*. No within-window median normalisation is applied. The dollar level is matched by the model's aggregate production scale A , which is estimated. Letting A float is the identified analogue of a free production scale (Lise–Meghir–Robin 2016): the overall wage level identifies A , the premium identifies γ_{P_S}/P_U , and the skilled wage curvature identifies the exponent γ_{P_S} , so A is separated from the relative productivities.

Measurement error. The observed log wage is the structural log wage plus mean-zero Gaussian noise, $\log w^{\text{obs}} = \log w^{\text{struct}} + \sigma_{w,j} \varepsilon$, $\varepsilon \sim N(0, 1)$, $j \in \{U, S\}$. Consequently the observed log-wage *variance* exceeds the structural variance by $\sigma_{w,j}^2$, the observed *percentiles* are the $\sigma_{w,j}$ -convolved quantiles, and the observed *mean* and *third central moment* are unchanged (symmetric noise). $\sigma_{w,j}$ is a model-side parameter; the data moments are computed directly on observed wages and carry whatever measurement error the survey contains.

Within-window aggregation: per-year-then-average. For all wage moments we compute the moment within each ASEC year in the window and then average across years. This puts every ASEC year on equal footing regardless of sample size, and prevents a single large-sample year from dominating.

mean_wage_U, mean_wage_S. ASECWT-weighted mean of $\log w$ by skill group, computed within each ASEC year, then averaged across years in the window.

p25_wage_U, p25_wage_S. ASECWT-weighted 25th percentile of $\log w$ by skill group, computed within each ASEC year, then averaged across years in the window.

p50_wage_U, p50_wage_S. ASECWT-weighted median of $\log w$ by skill group, computed within each ASEC year, then averaged across years in the window.

wage_premium. $E[\log w \mid S] - E[\log w \mid U]$, computed within each ASEC year and averaged. Invariant to the overall wage scale A and to symmetric measurement error.

overlap_UgtS, overlap_SltU (cross-market wage overlap). Computed within each ASEC year against that year’s skill-specific log-wage medians, then averaged across years:

$$\text{overlap_UgtS}_y = \frac{\sum_{i \in U_y: \log w_i > \text{med}_S} \text{ASECWT}_i}{\sum_{i \in U_y} \text{ASECWT}_i}, \quad \text{overlap_SltU}_y = \frac{\sum_{i \in S_y: \log w_i < \text{med}_U} \text{ASECWT}_i}{\sum_{i \in S_y} \text{ASECWT}_i},$$

where med_S (med_U) is the ASECWT-weighted median log wage of the skilled (unskilled) employed in year y . Both require both skill groups present in the year; they summarise where the two log-wage distributions sit relative to each other beyond the gap in means (**wage_premium**). Like the model counterpart (the $\sigma_{w,j}$ -convolved tail probability at the other segment’s median), they are computed on observed log wages.

Level check at the model-data interface. The data are in real weekly-earnings levels (logged); the model solver produces wages on its own internal scale, which the estimated A aligns to the data level. A one-time visual check of the empirical and model-implied log-wage densities, plotted on a shared axis, is sufficient to verify that the estimated A has brought the two level distributions into register.

13 Market tightness

Step 1. Unemployment by skill from CPS Basic: $U_{j,t}$.

Step 2. Vacancies by skill via industry allocation (Section 4). Let $s_{k,w}$ be the within-window weighted skilled share of employment in supersector k (computed from CPS using the industry-to-supersector map, with the working-student exclusion inherited from the **skilled_share** construction). Then

$$V_{S,t} = \sum_k s_{k,w} V_{k,t}, \quad V_{U,t} = \sum_k (1 - s_{k,w}) V_{k,t},$$

summing only over the 17 allocation supersectors (never the six aggregate cross-check series). JOLTS units are scaled by 1000 to match CPS counts.

Step 3. $\theta_{j,t} = V_{j,t}/U_{j,t}$; average over months in the window.

Part V SMM Weights, Influence Functions, and File Layout

14 SMM weight matrix

The SMM objective for window w is

$$Q_w(\theta) = [\hat{m}_w - m(\theta)]' \hat{\Xi}_w [\hat{m}_w - m(\theta)],$$

with $\hat{m}_w$ the vector of $K = 26$ data moments and $\hat{\Xi}_w$ a positive-semidefinite weight matrix. The headline estimation uses identity weights ($\hat{\Xi}_w = I$), equivalent to equal weight on each scale-normalised moment deviation. Two robustness configurations are available without re-running the data pipeline: diagonal weights $\hat{\Xi}_w = \text{diag}(1/\hat{\sigma}_{1,w}^2, \dots, 1/\hat{\sigma}_{K,w}^2)$, and the full optimal weight matrix $\hat{\Xi}_w = \hat{\Sigma}_w^{-1}$ (with off-diagonal shrinkage if the condition number of $\hat{\Sigma}_w$ exceeds a target).

Pipeline / SMM separation of concerns. The pipeline produces and stores the full 26×26 covariance matrix $\hat{\Sigma}_w$ for each window (with the `training_share` row/column already κ_w -adjusted in Stage 8). Weight-matrix construction is handled by the SMM code, which inverts (or diagonalises) $\hat{\Sigma}_w$ at load time. Keeping the full $\hat{\Sigma}_w$ on disk means the weight scheme can be changed without re-running the pipeline. The κ_w rescaling of `training_share` is likewise applied *in the pipeline*: Stage 7 writes the κ_w -adjusted moment to `moments_{window}.csv` and Stage 8 writes the κ_w -adjusted row/column into `sigma_{window}.csv` (see Section 15). The SMM driver reads both pre-adjusted.

15 Influence functions

Each moment $\hat{m}_k = \frac{1}{N} \sum_i \psi_k(z_i) + o_p(N^{-1/2})$. The variance-covariance matrix is $\hat{\Sigma}_w = \frac{1}{N_w} \sum_i \psi(z_i) \psi(z_i)'$.

Sample mean. $\psi_i = y_i - \hat{m}$. Covers `mean_wage_j` (with $y_i = \log w_i$).

Sample variance. $\psi_i = (y_i - \bar{y})^2 - \hat{\sigma}^2$. Covers `emp_var_j`.

Third central moment. $\psi_i = (y_i - \bar{y})^3 - \hat{\mu}_3$, the third-moment analogue of the variance case. Covers `emp_cm3_j`.

Ratio / rate. $\psi_i = (y_i - \hat{r} x_i) / \bar{x}$. This covers `ur_total`, `ur_j`, `skilled_share`, `training_share`, the CPS transition rates (`jfr_j`, `sep_rate_j`), `ee_rate_S`, `theta_j`, and the skilled long-term-unemployment share `ltu_share_S`.

Median / quantile. $\psi_i = (\mathbf{1}\{y_i \leq \hat{q}_\alpha\} - \alpha) / f(\hat{q}_\alpha)$ for the α -quantile. Covers `p25_wage_j` ($\alpha = 0.25$) and `p50_wage_j` ($\alpha = 0.50$).

Cross-market overlap (two-sample share). `overlap_UgtS` and `overlap_SltU` are weighted shares of one segment evaluated at the *other* segment's estimated log-median, so a fully efficient ψ carries an extra term in the estimated median (a two-sample influence function). The pipeline stacks all three appended moments (`ltu_share_S` monthly, the overlaps at ASEC-year frequency) with the common per-period-deviation convention $\psi = (\text{value}_{\text{period}} - \hat{m})$, so every moment shares one variance construction. This is exact under equal weights and a conservative approximation under the optimal-weight matrix; the analytic two-sample / ratio influence functions are documented here but not used in the stored $\hat{\Sigma}_w$.

Implementation. For each window: (i) compute $\hat{m}_w$; (ii) compute $\psi(z_i)$ for each observation; (iii) form $\hat{\Sigma}_w$; (iv) store both $\hat{m}_w$ and $\hat{\Sigma}_w$. When moments are aggregated across periods (e.g. ASEC year-then-window or CPS month-then-window), the influence-function contributions are stacked at the appropriate aggregation level before forming the outer product, so that within-period clustering is respected.

training_share rescaling (data pipeline, Stage 8). The $\hat{\Sigma}_w$ written to `sigma_{window}.csv` already carries the κ_w level adjustment on the `training_share` row and column: Stage 8 of the pipeline multiplies that row and column by κ_w (so the diagonal cell is multiplied by κ_w^2 and every off-diagonal entry in that row/column is multiplied by κ_w) before writing the file. Algebraically:

$$\hat{\Sigma}_w^{\text{adj}}[k, k'] = \begin{cases} \kappa_w^2 \hat{\Sigma}_w[k, k] & \text{if } k = k' = \text{training_share}, \\ \kappa_w \hat{\Sigma}_w[k, k'] & \text{if exactly one of } k, k' = \text{training_share}, \\ \hat{\Sigma}_w[k, k'] & \text{otherwise.} \end{cases}$$

This is exact under A1–A3 of Section 10. All other rows and columns are unchanged; the full CPS covariance structure of the other 25 moments is preserved. The SMM driver reads `sigma_{window}.csv` as written and applies no further rescaling.

16 File and directory layout

```
project_root/
  code/
    solver/                                     # model solver
      model_main.jl
      params.jl  grids.jl  unskilled.jl  skilled.jl
      solver.jl  equilibrium.jl          # equilibrium.jl returns ltu_share_S
      plots.jl  plots_main.jl  single_run_plots.jl
    smm/                                         # SMM estimation
      smm_main.jl                               # driver
      moments.jl                               # data + model moment loaders
                                              # (read the kappa-pre-adjusted
                                              #   moments_*.csv / sigma_*.csv)
      smm_params.jl                           # SMMSpec, SMMRunParams, free-param list
      smm.jl                                  # SMM objective, DE, SA, NM
      candidates.jl                           # Sobol + clustering candidate seed bank
      moment_covariance.jl                    # moment-redundancy / drop-list diagnostic
    data_processing/                            # data pipeline (run data_processing_main.jl)
      data_processing_main.jl                  # driver: paths, includes, runs stages 0-11
      setup.jl                                # WINDOWS, MOMENT_NAMES (26), helpers
      cps_basic.jl  cps_asec.jl                # CPS Basic / ASEC cleaning (DURUNEMP kept)
      jolts.jl      j2j.jl                     # JOLTS openings / J2J EE rates
      nsc.jl         # NSC enrolment: kappa_w + phi calibration
      transitions.jl # worker-flow hazards + nu life-table
      moments.jl     # 26 moments per window; writes the
                      #   kappa_w-adjusted training_share to
                      #   moments_*.csv (Stage 7)
      sigma.jl       # 26x26 Sigma; writes the kappa_w-adjusted
                      #   training_share row/col to
                      #   sigma_*.csv (Stage 8)
      validation.jl  # end-of-run diagnostics
    data/
      raw/
        cps_basic/      # IPUMS extract (incl. DURUNEMP)
        cps_asec/       # IPUMS ASEC extract
        jolts/          # BLS flat files
        j2j/            # Census J2J flows
        nsc/            # NSC Enrollment-Coverage Excel
      derived/
        windows.json    # WINDOWS dict; single source of truth
        cps_basic_clean.arrow
        cps_asec_clean.arrow
        transitions_monthly.arrow
        jolts_clean.arrow
        j2j_ee_rates.csv # J2J E4-only EE rates by window
        phi_calibration.csv # single phi pooled across Fall semesters
```

```

nu_estimation.csv          # life-table nu on base_fc and base_covid
training_share_scale.csv   # kappa_w per window (NSC/CPS level anchor)
cps_vs_nsc_enrollment.csv  # diagnostic table backing kappa_w
moments_base_fc.csv        # 26 moments per window
moments_crisis_fc.csv
moments_base_covid.csv
moments_crisis_covid.csv
sigma_base_fc.csv          # full 26x26 var-cov
                             # (training_share row/col already x kappa)

sigma_crisis_fc.csv
sigma_base_covid.csv
sigma_crisis_covid.csv
moment_scales_base_fc.csv  # per-moment scale factors (IF normalisation)
moment_scales_crisis_fc.csv
moment_scales_base_covid.csv
moment_scales_crisis_covid.csv

output/
  plots/
  tables/
  smm/                      # serialised SMMResult bundles (.jls)

```

17 Summary: moment-to-data mapping

Table 5: Moment-to-data mapping.

Moment	Source	Key variables
ur_total	CPS Basic	EMPSTAT, LABFORCE, SCHLCOLL, EDUC, WTFINL
ur_U	CPS Basic	same
ur_S	CPS Basic	EMPSTAT, LABFORCE, EDUC, WTFINL
skilled_share	CPS Basic	LABFORCE, SCHLCOLL, EDUC, WTFINL
training_share	CPS Basic + NSC	LABFORCE, SCHLCOLL, EDUC, AGE, WTFINL; NSC IPEDS-Universe Fall headcount for κ_w
emp_var_U	CPS ASEC	INCWAGE, WKSWORK1, UHRSWORKLY, EDUC
emp_cm3_U	CPS ASEC	same
emp_var_S	CPS ASEC	same
emp_cm3_S	CPS ASEC	same
jfr_U	CPS Basic (matched)	CPSIDP, EMPSTAT, EDUC
jfr_S	CPS Basic (matched)	same
sep_rate_U	CPS Basic (matched)	same
sep_rate_S	CPS Basic (matched)	same
ee_rate_S	Census J2J	EEHire, MainB, education = E4
mean_wage_U	CPS ASEC	INCWAGE, WKSWORK1, UHRSWORKLY, EDUC
mean_wage_S	CPS ASEC	same
p25_wage_U	CPS ASEC	same
p25_wage_S	CPS ASEC	same
p50_wage_U	CPS ASEC	same
p50_wage_S	CPS ASEC	same

Moment	Source	Key variables
wage_premium	CPS ASEC	same
theta_U	JOLTS + CPS	openings by industry, IND, EDUC
theta_S	JOLTS + CPS	same
ltu_share_S	CPS Basic	EMPSTAT, EDUC, DURUNEMP, WTFINL (model side also uses ν)
overlap_UgtS	CPS ASEC	INCWAGE, WKSWORK1, UHRSWORKLY, EDUC
overlap_SltU	CPS ASEC	same

Part VI Externally Set Parameters

18 Overview

Three parameters are set outside the SMM estimation loop. One is *calibrated* to a standard external value; two are *pre-estimated* from external micro-data. r and ϕ are common across all four windows; ν takes two values, one per crisis pair. The aggregate production scale A is *not* externally set: it is estimated inside the SMM loop, and is identified from the overall wage level as described in Section 8.

Table 6: Externally set parameters.

Symbol	Name	Value	Frequency	Source / method
<i>Calibrated, common across windows</i>				
r	Discount rate	0.00417	monthly	5% annual real rate (risk-free rate)
<i>Pre-estimated from data, common across windows</i>				
ϕ	Training completion rate	— (from data)	monthly	NSC/IPEDS Universe enrollment pooled across Fall semesters 2003–2024 + NCES median time-to-degree; enrollment-weighted across 4-year and 2-year programmes.
<i>Pre-estimated from data, per baseline window</i>				
ν_w	Demographic turnover	— (from data)	monthly	Life-table on CPS Basic LF (ages 16–64), $\hat{\nu}_w = 1/\overline{\text{remaining months}}$, computed on <code>base_fc</code> and <code>base_covid</code> . FC pair uses $\hat{\nu}_{\text{base_fc}}$; COVID pair uses $\hat{\nu}_{\text{base_covid}}$.

19 Calibrated parameter: discount rate r

The discount rate enters every value function as a level shifter: for a generic HJB of the form $(r + \nu + \dots)V = \dots$, a higher r uniformly lowers continuation values. This effect is almost perfectly collinear with changes in the flow payoffs b_U, b_T, b_S and the productivity shifters P_U, γ_{P_S} : raising r by ϵ and simultaneously adjusting flow payoffs or productivity can leave all observable moments nearly unchanged. Formally, the Jacobian $\partial m / \partial r$ (evaluated at the baseline) is close to a linear combination of $\partial m / \partial b_U$, $\partial m / \partial b_S$, and $\partial m / \partial P_U$, making the Hessian of the SMM objective near-singular when r is included. The discount rate has no moment that

pins it down independently of flow payoffs. We calibrate $r = 0.0033$ monthly ($\approx 4\%$ annual), the standard convention.

20 Pre-estimated parameter: training completion rate ϕ

20.1 Why ϕ is not in the SMM objective

In principle, ϕ is jointly identified by the training share and the training flow rate. In steady state, $\text{training_share} \approx \text{training_flow}/(\phi + \nu)$, so given the two moments one could back out ϕ . In practice the training flow is poorly measured at monthly frequency in the CPS (small enrollment-state transitions are dominated by sampling noise), and small measurement error in the training share translates into large error in ϕ because the training share itself is small ($\approx 2\%$). More fundamentally, ϕ interacts tightly with the training cutoff $\bar{x}$: a model with high ϕ (fast completion) and a low cutoff (few entrants) can produce similar training stocks as a model with low ϕ and a higher cutoff. Without direct microdata on training-spell durations, the two are not separately identified from CPS moments alone. We compute ϕ directly from administrative data on enrollment and time-to-degree.

20.2 Construction recipe

Data inputs. Two ingredients, pooled across all available Fall semesters:

1. Average IPEDS Universe enrollment in 4-year institutions ($E_{4\text{yr}}$) and in 2-year institutions ($E_{2\text{yr}}$), from the NSC Enrollment Coverage Report. For each available Fall semester between 2003 and 2024 we read the US-level rows for “All 4-year Institutions” and “All 2-year Institutions” and take the unweighted mean across years.
2. NCES median time-to-degree: $d_{4\text{yr}} = 49$ months (bachelor’s), $d_{2\text{yr}} = 37$ months (associate’s).

Formula. The enrollment-weighted average completion hazard across the two programme types is

$$\hat{\phi} = \frac{E_{4\text{yr}}(1/d_{4\text{yr}}) + E_{2\text{yr}}(1/d_{2\text{yr}})}{E_{4\text{yr}} + E_{2\text{yr}}}. \quad (2)$$

This is the monthly completion probability for an average enrollee. The same $\hat{\phi}$ is used in all four windows.

21 Pre-estimated parameter: demographic turnover ν_w

We estimate ν separately on each baseline window (`base_fc` and `base_covid`) using the life-table method: for each CPS Basic LF participant of age a , the implied remaining working life is $\max(65 - a, 0) \cdot 12$ months, and $\hat{\nu}_w$ is the reciprocal of its weighted mean over window w .

21.1 Why ν cannot be included in the SMM objective

The turnover rate ν appears in the model symmetrically with r : the effective discount rate in most HJBs is $r + \nu$. As a result, ν suffers from the same collinearity: a change in ν can be offset by adjusting r , leaving value functions and hence observable moments essentially unchanged. Beyond this collinearity, ν also governs the inflow rate of new workers (with density $\ell(x)$), which affects the stationary distributions. The distribution moments (`skilled_share`, `training_share`) are primarily driven by the training cutoff $\bar{x}$ and the type distribution (a_ℓ, b_ℓ), so the marginal effect of ν on these moments is swamped by the effects of the deep parameters the estimation already targets. Pre-estimating ν from a separate data source resolves this without imposing a single life-table assumption across the full sample.

21.2 Construction recipe: life-table on the LF

Data source. CPS Basic Monthly, same extract as for the transition moments (Section 2). Cross-section: no panel matching is required.

Sample.

1. Civilian, age 16–64.
2. In the labour force at date t (employed or unemployed, `LABFORCE` = 2).
3. Restricted to the window w being estimated ($w \in \{\text{base_fc}, \text{base_covid}\}$).

Remaining working life. For each LF observation (i, t) , the implied months remaining until age 65 are

$$\text{rem}_{i,t} = \max(65 - \text{AGE}_{i,t}, 0) \times 12.$$

The weighted mean over window w is

$$\overline{\text{rem}}_w = \frac{\sum_{(i,t) \in w} w_{i,t} \text{rem}_{i,t}}{\sum_{(i,t) \in w} w_{i,t}},$$

and the life-table turnover rate is the reciprocal:

$$\hat{\nu}_w = \frac{1}{\overline{\text{rem}}_w}.$$

Two values, one per crisis pair. We compute $\hat{\nu}_w$ for $w \in \{\text{base_fc}, \text{base_covid}\}$, and use the FC value $\hat{\nu}_{\text{base_fc}}$ for the (base_fc, crisis_fc) pair and the COVID value $\hat{\nu}_{\text{base_covid}}$ for the (base_covid, crisis_covid) pair. Allowing the two pairs to have distinct $\hat{\nu}$ values accommodates secular drift in the working-age composition of the LF between the early 2000s and the late 2010s (the LF has aged, lengthening $\overline{\text{rem}}_w$ slightly between the two baselines).

Why baseline windows. The pre-FC and pre-COVID baseline expansions are the “normal” regimes within each crisis pair. Computing the life-table within the baseline windows avoids contamination from crisis-period sample-composition shifts (e.g. disproportionate exit of older workers during downturns) and keeps the macro ν aligned with the data the SMM treats as the long-run regime for each pair.

Relation to the model. In the model, workers exit at Poisson rate ν regardless of labour-force status, and each exit is replaced by a newborn entrant drawn from $\ell(x)$. The life-table value $1/\overline{\text{rem}}_w$ is the right level for this: $\overline{\text{rem}}_w$ is the average months until forced retirement at age 65, so $\hat{\nu}_w$ is the average monthly hazard needed to cycle every worker out by then.